

Weather & Climate

This is your guided version. It uses shorter sentences, word banks, and fill-in-the-blank notes to help you learn one step at a time. Read slowly. Fill in every blank. You can do this!



Georgia Standard S6E4 · Supported Reading

WHAT YOU'LL LEARN

01 Weather vs. Climate

03 Uneven Heating

05 Air Pressure, Fronts & Storms

02 Layers of the Atmosphere

04 How Wind Forms

06 Hurricanes



Name: _____

Period: _____

Date: _____

4.1 Weather vs. Climate

My Learning Goal

I can tell the difference between weather and climate.

These two words are different:

- **Weather** = what the air is doing **right now** (sunny, rainy, hot).
- **Climate** = the **average** weather over **many years**.

Easy Trick

Weather is your mood. Climate is your personality. Mood changes fast. Personality is your usual pattern.

Word Bank

weather

climate

right now

many years

Fill in the blanks:

- "It is raining today" describes _____.
- "Summers here are always hot" describes _____.
- Climate is measured over _____.

4.2 Layers of the Atmosphere

My Learning Goal

I can name the layer we live in and what the ozone layer does.

The **atmosphere** is the air around Earth. It has layers.

Layers of the Atmosphere (lowest to highest)



Figure 4.1 — The layers of the atmosphere. We live in the bottom layer, the troposphere.

Two important parts:

- The **ozone layer** blocks the Sun's harmful UV rays. It is like sunscreen for Earth.
- **Greenhouse gases** trap heat and keep Earth warm.

Fill in the blanks:

- We live in the layer called the _____. (Hint: weather happens here)
- The _____ layer protects us from the Sun's UV rays.

4.3 Uneven Heating

My Learning Goal

I can explain that land and water heat up at different speeds.

The Sun does **not** heat Earth evenly.

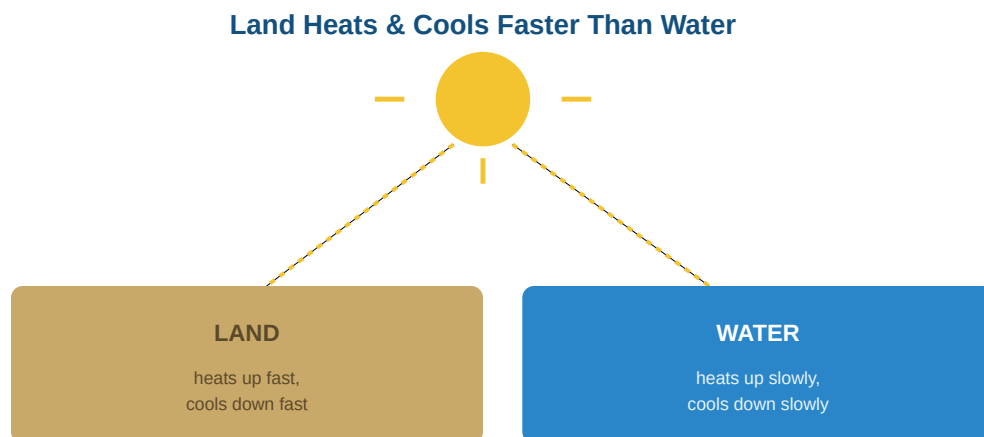


Figure 4.2 — Land heats up and cools down fast. Water heats up and cools down slowly.

Circle the correct answer:

- (Land / Water) heats up faster in the Sun.
- (Land / Water) keeps its heat longer at night.

★ Why It Matters

Uneven heating makes warm air and cool air sit next to each other. This is what makes **wind** blow.

4.4 How Wind Forms

🎯 My Learning Goal

I can explain how warm and cool air make wind.

Wind is moving air. Here is how it works:

- Warm air is light. It **rises**.
- Cool air is heavy. It **moves in** to take its place.
- That moving air **is** the wind.

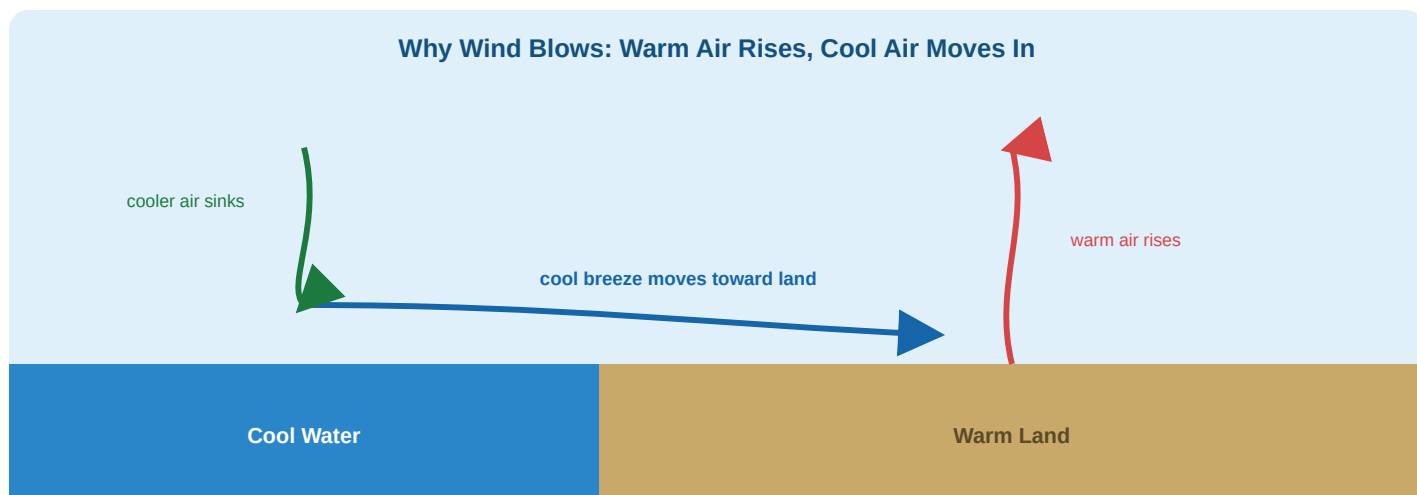


Figure 4.3 — Warm air rises over the land. Cool air blows in from the water. That is a sea breeze.

📝 Word Bank

rises

moves in

high

low

Fill in the blanks:

- When air gets warm, it _____.
- Then cooler air _____ to take its place.
- Wind flows from _____ pressure to _____ pressure.

4.5 Air Pressure, Fronts & Storms

My Learning Goal

I can describe what a front is and what storms it can cause.

An **air mass** is a big body of air. A **front** is where two different air masses meet. Storms often happen at fronts.

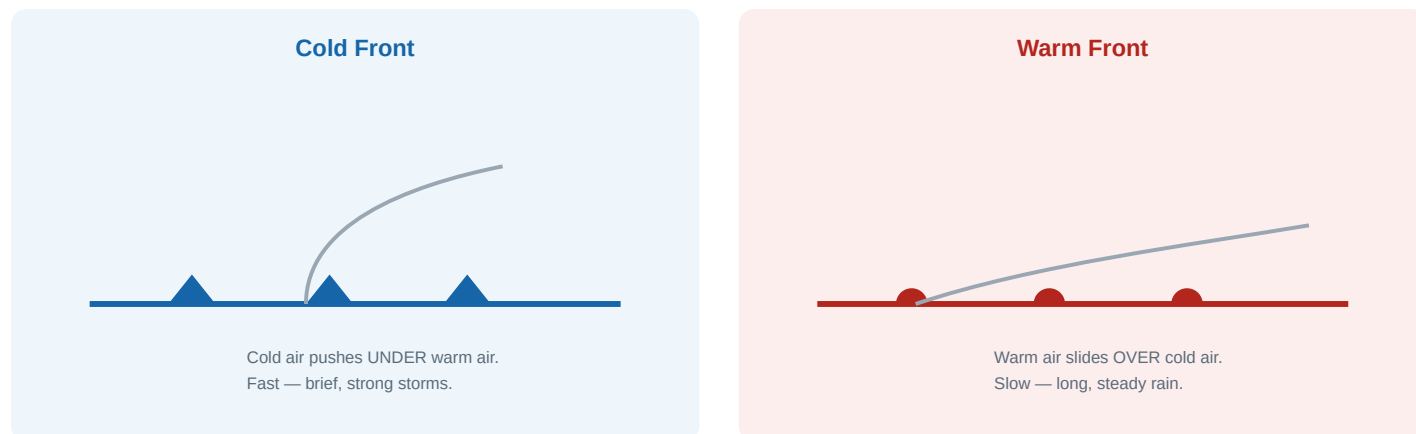


Figure 4.4 — A cold front (left) brings quick, strong storms. A warm front (right) brings slow, steady rain.

Front	What It Brings
Cold Front	_____ (Hint: sudden, strong)
Warm Front	_____ (Hint: slow, steady)

Remember

Low pressure = stormy weather. **High** pressure = clear weather.

A **thunderstorm** needs **warm** air and **moisture**. If the air also spins, it can make a **tornado**.

4.6 Hurricanes

My Learning Goal

I can explain where hurricanes get their energy.

A **hurricane** is the biggest, strongest storm on Earth.

It gets all its energy from **warm ocean water**.

How a Hurricane Forms Over Warm Ocean

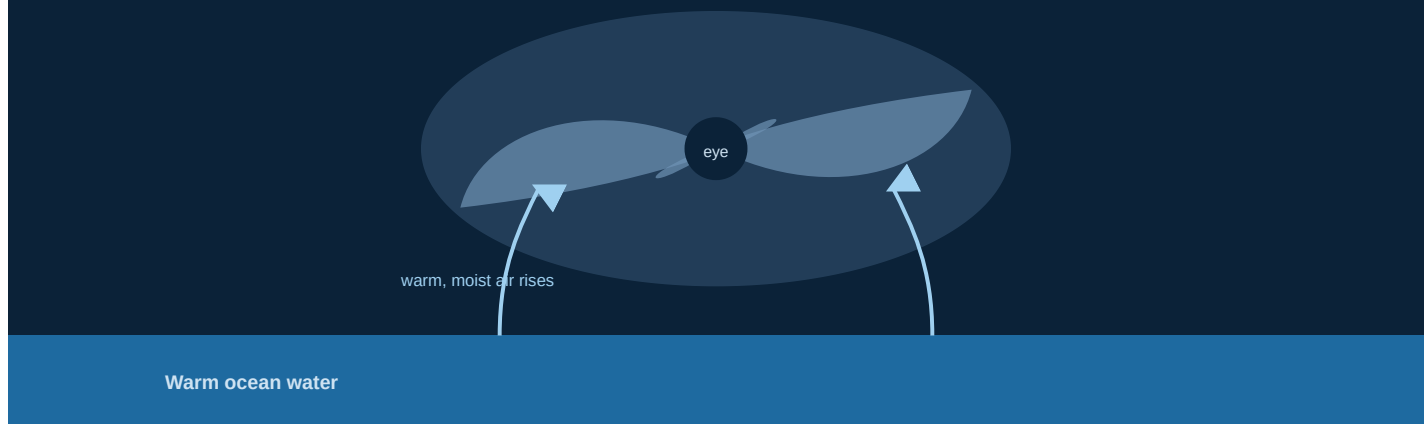


Figure 4.5 — Hurricanes form over warm ocean water. The calm center is called the "eye."

How it works:

- The Sun heats the ocean.
- Warm water **evaporates** and rises.
- This gives the storm energy to grow.
- Over land, there is no warm water, so the hurricane gets **weaker**.

Fill in the blanks:

- Hurricanes get their energy from _____ ocean water.
- The calm center of a hurricane is called the _____.

Chapter 4 Review

? Show What You Know

1. "It is sunny today" describes _____ (weather / climate).
2. We live in the atmosphere layer called the _____.
3. _____ heats up faster than water in the Sun.
4. Wind flows from _____ pressure to _____ pressure.
5. Hurricanes get their energy from _____ ocean water.